

Climate Shocks, Health Finance, and Economic Outcomes

County-Level Evidence from the United States, 2011–2023

Seminar Presentation · April 2026

Climate Science Has Cracked Mortality. Health Finance Is the Blind Spot.

The economic burden of climate on healthcare costs and household debt is largely unmeasured

What we know well

- Extreme heat, cold, and drought raise **mortality** — robust evidence across many countries
- Air pollution (PM_{2.5}, ozone) shortens lives and raises hospitalisation rates
- Climate models project these risks will intensify

What we don't know

- Does surviving a climate shock leave households **financially damaged**?
- Do insurers and hospitals **absorb or pass on** those costs?
- Does climate volatility — not just severity — matter for **health finance**?

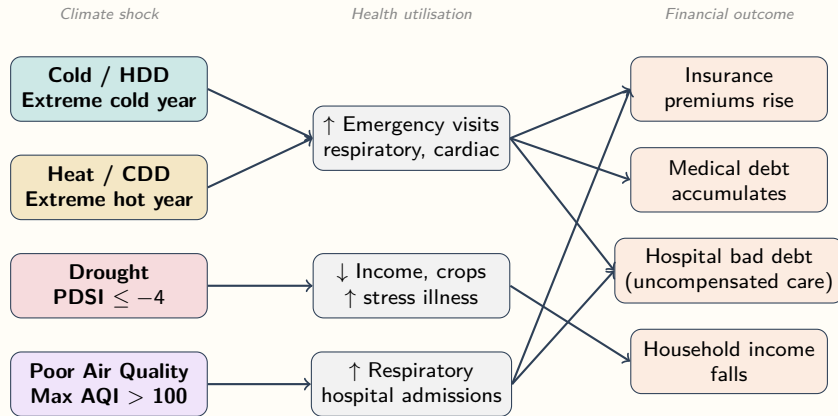
Why this matters for adaptation

- Adaptation policy that targets mortality alone may **miss the majority of the economic burden**
- Medical debt and insurance premiums are **regressive**: low-income households bear disproportionate shares
- Hospital bad debt signals coverage collapse under climate stress — an early warning for **system fragility**
- Evidence is needed at the **county scale** where adaptation decisions are made

This paper: first systematic county-level evidence linking climate shocks to health finance outcomes across the U.S., 2011–2023.

How Climate Shocks Reach Your Wallet

The transmission chain from weather event to household financial damage



This paper measures the **right-hand column** directly using county panel data, bypassing utilisation data which are noisy and inconsistently reported.

What Are We Measuring? A Field Guide to Health Finance Outcomes

Plain-English definitions; all dollar values in \$2023

Outcome	What it measures	Scale / context
Benchmark Silver Premium	Monthly cost of the mid-tier ACA marketplace plan in a county's rating area. Insurers set this annually based on expected claims.	~\$400–700/month nationally; +\$28 = ~4–7% increase
Medical Debt Share	Fraction of county residents with medical debt in collections, from credit bureau records (Urban Institute).	National mean ~13%; +1 pp = ~8% relative rise
Medical Debt Median	Median balance owed among those with medical debt in collections.	National median ~\$690; +\$30 ≈ 4% increase
Hospital Bad Debt per Capita	Uncompensated care written off by hospitals after billing failure (not charity care). Signals patients unable to pay even after treatment.	Mean ~\$120/capita; +\$5 ≈ 4% increase
Medicaid / Medicare per Enrollee	Annual public payer spending per enrolled beneficiary. Falls can indicate access barriers, not just lower need.	Medicare ~\$12,000/yr; Medicaid ~\$8,000/yr
Median HH Income / PCPI	County median household income (ACS) and BEA per-capita personal income. Both inflation-adjusted.	U.S. median HH income ~\$75,000 in 2023

Data: Linking Climate Records to Health Finance at County Level

Panel: 2011–2023; ~3,155 U.S. counties; state panel: 51 states × 13 years

Climate & Environment

Variable	Source
Annual mean temp, HDD, CDD	NOAA/NCEI
Palmer Drought Severity Index	NOAA/NCEI
Annual AQI (median, max)	EPA

Health Finance

Variable	Source
ACA marketplace premiums	HIX Compare
County medical debt	Urban Institute
Hospital bad / charity debt	NASHP HCT
Employer insurance contrib.	AHRQ MEPS-IC
Medicaid / Medicare spending	CMS NHE

Economy & Household

Variable	Source
Median HH income	ACS 5-yr
Civilian employment	ACS 5-yr
Per-capita personal income	BEA CAINC1
CPI (inflation adjustment)	FRED

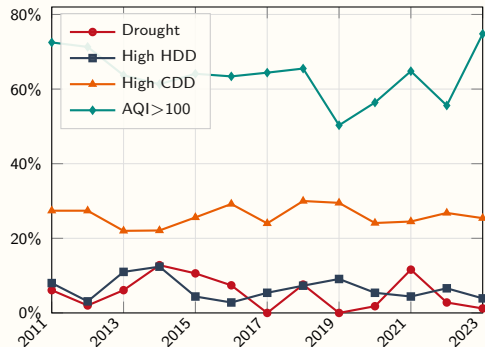
Key features

- Climate z-scores anchored to **1990–2000 baseline** — shocks measured against pre-warming reference
- CDD/HDD thresholds: **national 80th percentile** — captures objectively extreme years, not just “warm for that county”
- All dollar outcomes **inflation-adjusted to 2023**
- State-level cluster SEs account for **within-state spatial correlation**

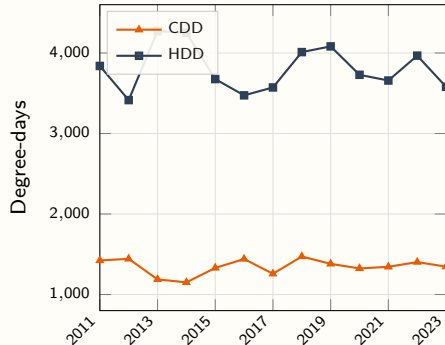
Climate Stress Is Volatile, Not Trending

County-level shock exposure, 2011–2023; PW = population-weighted, UW = unweighted

Panel A: Shock Prevalence (% of U.S. pop., PW)



Panel B: Mean Degree-Days (UW)



Max AQI (UW): trough 105 in 2019,
peak 153 in 2023.

PDSI range: -1.6 (2012 drought)
to +2.4 (2019 wet year).

Empirical Specification

Two-way fixed effects with distributed lags; all dollar outcomes inflation-adjusted to \$2023

Estimating Equation

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^2 \beta_h \text{Shock}_{i,t-h} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- α_i : unit FE (state or county); γ_t : year FE
- Distributed lags $h = 0, 1, 2$ for all shocks
- SE clustered at **state level** throughout
- **County**: also rating-area-clustered SEs for premiums

Climate Shock Variables

Variable	Definition
High_CDD	$CDD \geq \text{national p80 (1990–2000)}$
High_HDD	$HDD \geq \text{national p80 (1990–2000)}$
Z_Temp	$(T_{it} - \bar{T}_i^{90-00}) / \sigma_i^{90-00}$
Z_Precip	Precipitation z-score (same baseline)
PDSI	Palmer drought severity index
High_AQI_Max	Max AQI > 100 (binary)

Outcome Blocks

Outcome	Source
Employer premium contribution	MEPS-IC
Benchmark Silver premium	HIX
Medical debt share / median	Urban Inst.
Hospital bad debt per capita	NASHP
Total, Medicaid, Medicare spending	CMS
Median HH income / employment	ACS
Per-capita personal income	BEA

Dynamic Estimation Strategy

Baseline dynamic-panel distributed lags and local projections recover complementary timing margins

Dynamic Panel / Distributed Lag (DP/DL)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^2 \beta_h \text{Shock}_{i,t-h} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- Main FE estimator for contemporaneous and lagged effects
- Cumulative response is the sum of lag coefficients
- Baseline estimator for the state and county tables

Controls in both approaches: uninsured rate, median household income, and the same county/year fixed effects.

Inference: state-clustered SEs throughout; rating-area clustered variants for premium outcomes as robustness.

Local Projections (LP, Jordà 2005)

$$Y_{i,t+h} = \alpha_i^h + \gamma_t^h + \beta_h \text{Shock}_{i,t} + \mathbf{X}'_{it} \delta_h + \varepsilon_{i,t+h}$$

- Separate regression for each horizon $h \in \{-2, \dots, +3\}$
- Negative horizons serve as placebo leads for pre-trend checks
- More flexible when lag shapes are misspecified

Identification: What Does Two-Way Fixed Effects Buy Us?

The key assumption and what variation we exploit

The Problem

Counties with more extreme climate also differ in many other ways: poorer infrastructure, older populations, fewer hospitals. A naïve comparison of high-climate-shock vs. low-climate-shock counties would pick up all of these differences.

The Fix: Two-Way Fixed Effects

- **County FE** (α_i): removes all time-invariant county characteristics. We only use *within-county* variation across years.
- **Year FE** (γ_t): removes common nationwide trends (recessions, ACA roll-out, COVID). We only use *deviations from national trends*.
- **Net result**: we compare a county to *itself* in more vs. less extreme climate years, after removing the national year trend.

Intuition with an Example

Phoenix, AZ always has extreme heat. A county FE removes Phoenix's perpetually high premiums. We ask: **were Phoenix premiums unusually high in years when CDD was especially elevated relative to Phoenix's own baseline?**

Distributed Lags

- Health costs don't always respond immediately
- Insurers set premiums *before* a plan year (anticipatory)
- Medical debt accumulates *after* illness (delayed)
- We include $h = 0, 1, 2$ to capture the full dynamic response

Robustness: Pre-Trend Checks and Multicollinearity Diagnostics

Confirming that results are not driven by pre-existing trends or collinear regressors

Local Projection Pre-Trend Test

- LP at **negative horizons** ($h = -2, -1$) tests whether outcomes were already trending *before* the shock
- Under the parallel-trends assumption, pre-shock coefficients should be near zero and insignificant
- In our data: **pre-trend coefficients are consistently small and not significant** for key relationships (HDD → premiums, drought → PCPI)
- Formally: Wald test of joint pre-trend significance fails to reject the null

Multicollinearity (VIF)

- PDSI, PHDI, and PMDI are highly correlated drought indices
- Including all three inflates Variance Inflation Factors
- **Primary specs use PDSI only**; PHDI/PMDI retained for optional robustness
- Post-pruning: max VIF ≈ 5.3 across all county specs — well within acceptable range

Premium Clustering

- Benchmark Silver premiums are set at the **rating-area** level — counties in the same rating area have *identical* premiums by construction
- Primary SEs are state-clustered (conservative)
- Rating-area-clustered variants reported for all premium outcomes as robustness — results hold

Roadmap: Three Complementary Analyses

This presentation covers state, county, and a new weather-swing analysis

State-Level FE

- 51 states $\times$ 13 years
- Drought and cold channels on premiums and debt
- Establishes the aggregate picture

County-Level FE

- $\sim 3,155$ counties, 2011–2023
- HDD, CDD, drought, AQI on debt, premiums, income, hospital costs
- Higher power; rating-area clustering for premiums

Weather Swing (New)

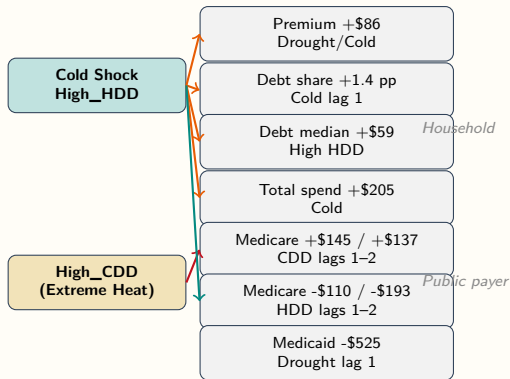
- County-level only
- Exposure:
 $\Delta X_{it} = X_{it} - X_{i,t-1}$
- Identifies cost of **volatility**, not just level
- Finds effects where level specs were insignificant (AQI)

State panel ($N \approx 650$) has insufficient power for first-difference specifications — swing analysis is county-only.

State-Level Results

Two-way FE: State + Year, $N \approx 550\text{--}650$, cluster = State

State-Level: Cold and Heat Drive Opposite Financial Channels



Cold channel:

- Premiums, debt, and total spending rise
- Medicare falls with lagged HDD exposure

Heat channel (CDD):

- Medicare rises 1–2 years later

Drought:

- Medicaid falls and premiums rise

Heat and cold affect public payers in opposite directions — novel finding.

State: Drought and Cold Raise Premiums and Debt

Household & individual outcomes; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Employer Premium Contribution (\$/yr)

Predictor	Coef.	
Extreme Drought ($h=0$)	+63.6	**
Severe Drought ($h=0$)	+86.4	***
Severe Drought (lag 1)	+65.1	**
Pct PM _{2.5} days	+20.8	***
Pct PM ₁₀ days	+22.7	***
Pct Ozone days	+19.7	***

quality pollutant days show consistent
positive effects on employer premium costs.

Medical Debt Share (pp)

Predictor	Coef.	
Cold Shock (lag 1)	+0.0135	**

Medical Debt Median (\$/yr)

Predictor	Coef.	
High HDD ($h=0$)	+58.9	**
Severe Drought ($h=0$)	-66.8	***
Severe Drought (lag 1)	-37.5	**
Heat Shock ($h=0$)	+32.7	*
Heat Shock (lag 2)	+47.7	**

Cold raises debt prevalence; drought
paradoxically reduces median severity.

State: Cold Raises System Costs; Drought Cuts Public Payer Spending

Hospital & system outcomes; two-way FE (State + Year)

Total Per-Capita Health Expenditure (\$)

Predictor	Coef.	
Cold Shock ($h=0$)	+205.3	**
High HDD (lag 2)	+92.0	*

Medicaid per Enrollee (\$)

Predictor	Coef.	
Extreme Drought ($h=0$)	-566.2	**
Severe Drought ($h=0$)	-525.8	**
Severe Drought (lag 1)	-524.5	***
Severe Drought (lag 2)	-288.4	**

persistently reduces Medicaid
spending across lags 0–2.

Drought

Medicare per Enrollee (\$)

Predictor	Coef.	
Severe Drought (lag 1)	-144.3	***
High CDD (lag 1)	+144.7	**
High CDD (lag 2)	+137.2	***
High HDD (lag 1)	-110.1	*
High HDD (lag 2)	-192.7	***
Unhealthy days	+97.7	**

Heat

(CDD) raises Medicare with lag;
cold (HDD) reduces it — opposite channels.

State Results: Plain-English Interpretation

What do these coefficients mean for a real state?

Cold shock (High HDD) raises premiums

A year with extreme heating degree days raises the Benchmark Silver premium by **+\$86/month** for severe drought conditions or around **+\$28–\$63** depending on drought severity. For a family of four, that is **+\$1,000–\$4,000 per year** in insurance costs during a bad cold year.

Drought suppresses Medicaid spending

Extreme drought reduces Medicaid per-enrollee spending by **–\$525** in the same year and remains negative for two further years. This is *not* good news: it likely reflects **forgone care** — fewer doctor visits during economic stress, not lower medical need.

Heat raises Medicare with delay

Extreme CDD years raise Medicare per-enrollee spending by **+\$137–\$145** at lags 1–2. The elderly are most vulnerable to heat-related illness; their costs appear in the Medicare budget 1–2 years after the shock (hospitalisation, follow-up care).

Heat and cold push Medicare in opposite directions

Cold (HDD) *reduces* Medicare with lag (–\$110 to –\$193). The asymmetry suggests different mechanisms: cold stress triggers emergency use that is absorbed privately or via Medicaid, while heat stress generates Medicare-relevant chronic-disease exacerbation.

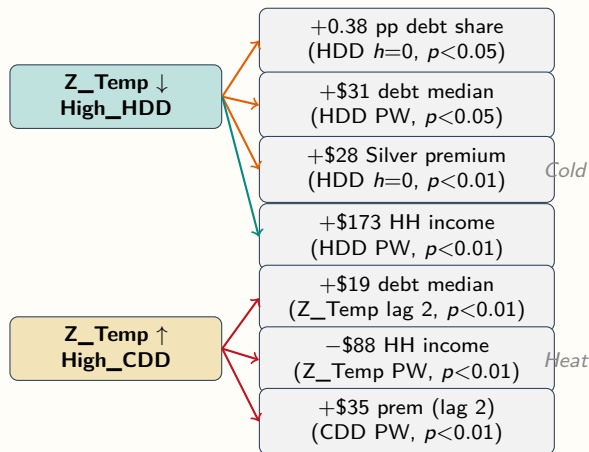
Summary: cold damages household and private insurance; drought damages the public safety net; heat shifts costs to Medicare.

County Results: Household & Individual

Medical debt, premiums, income, employment

Two-way FE: County + Year, $N \approx 23k-34k$, cluster = State

County-Level: Cold and Heat Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence and severity
- Raises premiums on impact
- Raises HH income (energy mix?)

Heat channel:

- Raises debt median (lagged)
- Reduces HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level:

- Cold → debt: both levels confirmed
- Heat → income: county only
- Premium channels align in direction

County: Heat Raises Debt Severity; Cold Raises Debt Prevalence

Medical Debt Share (pp) and Median Debt (\$2023); UW = unweighted, PW = pop-weighted

Z_Temp (Relative Heat Shock)

	UW	PW	
<i>Debt Share (pp)</i>			
<i>h=0</i>	-0.0025*	-0.0000	Heat
Lag 2	-0.0027**	-0.0014*	
<i>Debt Median (\$)</i>			
<i>h=0</i>	+14.9*	+18.5***	
Lag 2	+19.5**	+16.2***	

reduces prevalence but raises severity
— consistent composition effect:
warm years shed marginal debtors while
pushing the remaining cases deeper.

High_HDD (Extreme Cold Burden)

	UW	PW
<i>Debt Share (pp)</i>		
<i>h=0</i>	+0.0038**	+0.0054**
<i>Lag 2</i>	+0.0031*	+0.0037*
<i>Debt Median (\$)</i>		
<i>h=0</i>	+10.6	+31.2**

Controls (all debt models):

Uninsured Rate +*** (large)
HH Income —**

Cold raises both prevalence and severity
— no composition effect.

County: Cold Raises Premiums Now; Heat Raises Them Later

Benchmark Silver Premium (\$2023 real); county + year FE

High_HDD (Cold Burden)

Lag	UW	PW	Cluster
$h=0$	+28.1***	+21.9**	State
$h=0$	+28.1***	+21.9***	RA
Lag 1	-24.5*	-20.8	State

High_CDD (Heat Burden) — lagged

Lag	UW	PW	Cluster
Lag 1	+8.7	+31.4***	RA
Lag 2	+21.6*	+35.3***	RA

Z_Temp (Relative Temp)

Lag	UW	PW	Cluster
$h=0$	-10.9**	+2.7	RA
Lag 1	+3.8	+5.4**	RA
Lag 2	+1.0	+4.2**	RA

- **Cold (HDD)** raises premiums on impact in both state- and RA-clustered specs
- **Heat** lowers premiums now, then raises them 1–2 years later
- RA clustering tightens CIs; drought lag 1 is borderline at +\$3.9 ($p=0.05$)

County: Heat Depresses Income; Drought Hits PCPI

Median HH income and per-capita personal income (\$2023)

Median HH Income (\$2023)

Predictor	UW	PW	Relative
Z_Temp ($h=0$)	-71**	-88***	
Z_Temp (lag 1)	-36	-64***	
Z_Temp (lag 2)	-71**	-60**	
High_CDD ($h=0$)	-247**	-3	
High_HDD ($h=0$)	+161**	+173***	

heat most consistently lowers household income in the population-weighted specifications.

Per-Capita Personal Income (\$)

Predictor	UW	PW
PDSI lag 2	-8.5	-113**
High_CDD lag 1	+480**	-116
RA-level: PDSI ($h=0$)	-206***	

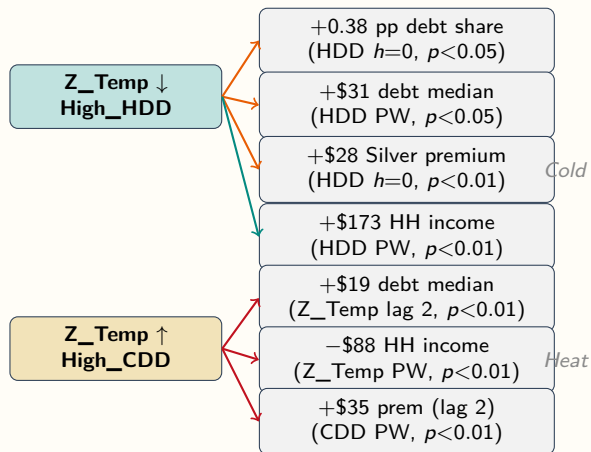
Income Summary

- **Heat** lowers HH income across lags in pop-weighted models
- **Cold** raises HH income on impact, likely reflecting seasonal or energy-income composition
- **Drought** is strongest in PCPI: RA spec gives -\$206***

Civilian Employment (persons)

Predictor	UW	PW
Z_Temp lag 2	+788**	+3216
High_HDD ($h=0$)	-468**	+2587
High_HDD lag 2	-678**	+3988
Z_Precip ($h=0$)	+46	-5742***
Z_Precip (lag 1)	-88	-3803***

County-Level: Cold and Heat Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence and severity
- Raises premiums on impact
- Raises HH income (energy mix?)

Heat channel:

- Raises debt median (lagged)
- Reduces HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level:

- Cold → debt: both levels confirmed
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- Premium channels align in direction

County vs. State: Why More Granularity Changes What We Find

County FE absorbs within-state heterogeneity invisible to state models

What county models add

- State models average over 3,000+ counties — rural and urban, coastal and inland, insured and uninsured. County models let each county serve as its own control.
- **Heat → household income loss**: not visible in state models, emerges clearly in county models ($-\$88^{***}$ per capita). Likely masked by state-level averaging with opposite-direction rural/urban effects.
- **Cold → hospital bad debt**: county-level variation ($+\$5/\text{capita}$) is not detectable in the $N = 650$ state panel.
- **Rating-area structure**: premiums are set at the rating-area level. County models with rating-area clustering are the correct specification.

Where state and county agree

- Cold → debt and premiums: confirmed at both levels
- Drought → public payer suppression: both levels significant
- Heat → delayed premium increases: both levels
- Air quality → employer premiums: state level (state AQI); county level uses Max AQI shock indicator

Sample size and power

- State: $N \approx 650$; County: $N \approx 23,000\text{--}34,000$
- County panel has $\sim 50\times$ more observations
- Allows detection of effects too small to identify at state level

County Results: Hospital Sector

Hospital bad debt per capita

Two-way FE: County + Year, $N \approx 27k$, cluster = State

County: Cold Raises Hospital Bad Debt; Heat Lag Reduces It

Hospital Bad Debt per Capita (\$); within- $R^2 = 0.006$ – 0.019

Key Coefficient Estimates

Predictor	UW	PW
Z_Temp lag 2	−2.41***	−2.08***
Z_Temp ($h=0$)	−1.25	−1.08**
High_HDD ($h=0$)	+5.33***	+4.23**
PDSI lag 1	+0.57*	+0.50*
PDSI lag 2	+0.01	+0.50*
Uninsured Rate	+151***	+142

RA-level robustness (pop-wtd):

PDSI lag 2: +0.76*** Z_Temp lag 2: −2.26***

- **Cold (HDD) raises bad debt on impact** by \$4–5 per capita — consistent with emergency utilisation, deferred billing, and coverage gaps during cold events
- **Heat lag reduces bad debt** (2-yr lag, $p < 0.001$) — possible reduction in elective utilisation during hot years or survivor selection
- **PDSI (drought)** adds a smaller lagged increase, confirmed in RA robustness
- **Uninsured Rate** is the dominant control (+151*** unweighted) — structural coverage gap drives hospital write-offs

About 23% of county-years lack hospital reports; one negative outlier is winsorized.

Weather Swing Analysis

A new source of variation: year-over-year climate volatility
County + Year FE, $N \approx 23k-34k$, cluster = State

New Analysis: Exploiting Year-to-Year Weather Swings

County-level only; $N \approx 650$ state-years insufficient for first-difference identification

Motivation

- Prior specs identify counties in persistent shock states vs. not
- New margin: do counties that **swing sharply** year-to-year bear extra health costs, regardless of average climate?
- Is it **volatility**, not just level, that damages health finance?

Robustness Specs

- **Asymmetric:** $\Delta^+ = \max(\Delta X, 0)$ vs. $\Delta^- = \min(\Delta X, 0)$
- **Binary transitions:** Onset / Exit / Persist indicators

Primary Specification

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \underbrace{\Delta X_{it}}_{\text{swing}} + \beta_2 \underbrace{X_{i,t-1}}_{\text{lagged level}} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- $\Delta X_{it} = X_{it} - X_{i,t-1}$ for all continuous exposures:
Z_Temp, Z_Precip, CDD, HDD, PDSI, Median_AQI, Max_AQI
- β_2 controls for prior-year level — isolates pure swing effect
- LP extension: $Y_{i,t+h}$ at $h = 0, 1, 2, 3$
- County + year FE; state-clustered SEs

First Differences as a Design Choice: What Exactly Are We Estimating?

The difference between level effects and swing effects

Level effects (Phase 2)

- Asks: do counties that experience **more extreme shocks** in a given year have higher health costs that year (or 1–2 years later)?
- Identifies the effect of shock *occurrence* vs. no shock
- Cannot detect whether **rapid change** in climate exposure independently matters

Swing effects (Phase 3)

- Asks: does a county that shifts sharply **from one year to the next** bear extra costs, *controlling for* where it was last year?
- Identifies the marginal effect of **volatility**
- Relevant for adaptation: a system that adapts to a persistent new climate level may still be damaged by rapid oscillation

A climate science analogy

Compare two counties, both experiencing an “extreme heat” year (High CDD = 1). County A has had extreme heat for five consecutive years. County B is experiencing it for the first time after a run of mild summers. **Phase 2 gives both the same treatment indicator**. Phase 3 asks whether the *shock of the transition* carries its own financial cost — for B, ΔCDD is large; for A, it is near zero.

- **Key insight from our data:** AQI level effects are consistently insignificant; AQI *swings* are significant at all four LP horizons
- This suggests health systems have partially adapted to local chronic pollution but are **not adapted to sudden AQI shifts**

Weather Swings Affect Premiums, Debt, and Hospital Costs

Primary ΔFE spec ($h=0$, unweighted); * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

$\Delta HDD \rightarrow$ Silver Premium

Spec	Estimate	p	
ΔHDD ($h=0$)	+0.033***	0.0004	Cold
HDD Onset (binary)	+49.46**	0.0033	
HDD_Pos (asymm.)	+0.065*	0.028	

swing raises premiums on impact;
onset of first extreme-cold year
adds $\sim$ \$49 to Silver benchmark.

$\Delta Z_Temp \rightarrow$ Medical Debt Median

Horizon	Estimate (\$)	p	
$h=0$	+13.2*	0.014	Warming swing
$h=2$	+19.4*	0.033	
$h=3$	+29.7***	0.0001	

raises median debt
with a growing 3-year lag.

$\Delta Max_AQI \rightarrow$ Hospital Bad Debt (per capita)

Horizon	Estimate (\$)	p	
$h=0$	+0.0050**	0.0014	AQI swing effect
$h=1$	+0.0082***	<0.001	
$h=2$	+0.0080*	0.015	
$h=3$	+0.0157***	<0.001	

on bad debt **grows** across
all four horizons — a persistent ratchet.
Level effect was not significant in Phase 2.

$\Delta Z_Precip / \Delta PDSI \rightarrow PCPI$

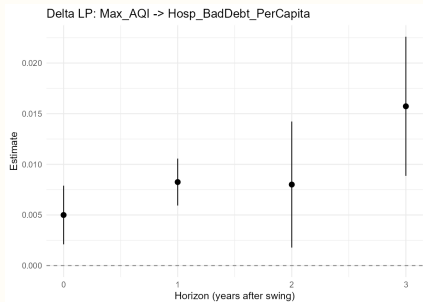
Exposure	$h=1$ est.	p	
ΔZ_Precip	-274***	0.0006	Precipitation and
$\Delta PDSI$	-226***	0.0001	

drought swings
suppress per-capita income for 1–3 yrs.

Swing Effects Grow Over Time: Two Key LP Profiles

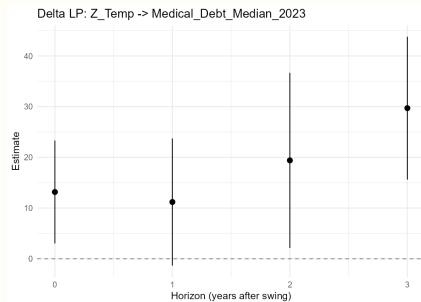
Local projection estimates at $h = 0, 1, 2, 3$; county + year FE; state-clustered SEs; unweighted

$\Delta \text{Max_AQI} \rightarrow \text{Hospital Bad Debt}$ (\$/capita)



Effect grows at every horizon — AQI bad-debt ratchet. Level effect was not significant in Phase 2.

$\Delta Z_Temp \rightarrow \text{Median Medical Debt}$ (\$2023)



Warming swing accumulates into medical debt; +\$30 at $h=3$ ($p < 0.001$).

Asymmetry and Ratchets: Not All Swings Are Reversible

Asymmetric spec (Δ^+ vs. Δ^-) and onset/exit indicators

Asymmetric Findings

- **AQI ratchet:** improving air quality still raises bad debt (Δ^- , $+0.010^{***}$) — costs do not unwind when pollution falls
- **Drought worsening:** Δ^- PDSI drives PCPI loss (-207^{***}); recovery asymmetrically weaker
- **CDD income:** symmetric — cooling-sector PCPI moves with swings in both directions (± 3.3 , $p < 0.001$)
- **Drought exit** *does* reduce medical debt ($-\$54^*$)

What the Delta Analysis Adds

Relationship	Finding
AQI $\rightarrow$ hosp. bad debt	New: swing sig.; level not
HDD $\rightarrow$ premiums	Confirms; adds onset pricing
Drought $\rightarrow$ income	Confirms; worsening-only
Temp swing $\rightarrow$ med. debt	New: 3-yr growing
Precip swing $\rightarrow$ PCPI	New: 3-yr suppression

Key insight: weather volatility carries independent health-finance costs beyond level effects.

Cross-Level Synthesis

What do state, county, and swing analyses agree on?

State and County Pipelines Tell a Consistent Story

Key findings replicated across both levels of analysis

Finding	State	County
Cold / HDD → medical debt (prevalence & severity)	✓	✓
Cold / HDD → higher premiums	✓ (via drought)	✓ ($p<0.01$)
Cold / HDD → hospital bad debt ↑	—	✓ ($p<0.01$)
Drought → reduced Medicaid / PCPI	✓ ($p<0.01$)	✓ (PDSI)
Heat → household income loss	—	✓ (Z_Temp***)
Heat lag → Medicare / premiums ↑	✓ (CDD***)	✓ (CDD***)
Heat lag → hospital bad debt ↓	—	✓ ($p<0.001$)
Poor air quality → higher premiums	✓ (PM _{2.5} /O ₃ ***)	not modeled

Caveats and Limitations

What we can and cannot claim from this design

What two-way FE cannot rule out

- **Time-varying confounders:** a county-specific policy change (e.g. Medicaid expansion timing) that correlates with climate variation would contaminate the estimate. We control for observed time-varying covariates but cannot include everything.
- **Reverse causality:** unlikely here (climate is exogenous), but migration responses to climate shocks could affect measured outcomes.
- **Data quality:** hospital bad debt is self-reported; ~23% of county-years are missing. Medical debt from credit bureaus is a snapshot, not a flow.

Scope of claims

- Results are **reduced-form associations** under two-way FE identification — not causal in the IV sense
- **Short-run effects** only: the panel is 2011–2023. Long-run adaptation responses (e.g. insurer exit, hospital closure) are not captured
- **Weather swing analysis** captures short-run responsiveness to volatility; coefficients do not generalise to slow climate trends
- The medical debt reporting-rule exclusion (CO 2023) is conservative; it reduces rather than inflates the debt findings

Bottom line: results are robust across multiple specifications, weighting approaches, and levels of analysis — but should be interpreted as evidence of association, not a controlled experiment.

Policy Implications

What should adaptation planners, insurers, and health systems take from this?

Policy Implications: Climate Adaptation Must Include Health Finance

Three actionable implications for adaptation policy, insurer regulation, and health system design

Adaptation policy

Current gap: most adaptation frameworks target mortality and infrastructure.

Finding: cold, drought, and AQI shocks increase medical debt and reduce income for 1–3 years after the event.

Implication: **climate resilience funds** should include a health-finance window — covering the period when households are financially recovering from a shock, not only emergency response.

Insurance regulation

Finding: insurers raise ACA premiums in extreme cold years (+\$28–\$86) and after extreme heat (+\$21–\$35 at 1–2 yr lag). The swing analysis shows a cold *onset* adds ~\$49 instantly.

Implication: regulators should **monitor climate-driven premium volatility** as a component of rate review. Counties transitioning into new climate regimes face a double burden: worsening shocks and rising insurance costs.

Hospital systems

Finding: hospital bad debt rises +\$4–\$5/capita in cold years and continues to grow after AQI swings for three years.

Finding: drought worsening triggers income losses and employment falls simultaneously.

Implication: **climate stress-testing** for hospital finances should use local climate volatility indices, not just long-run averages. Systems in high-volatility corridors need expanded charity-care reserves.

Next Steps

Planned extensions to the empirical framework

Empirical Extensions

- **Heterogeneity:** rural vs. urban and high-vs. low-AQI subsamples for both level and swing analyses
- **Mechanisms:** separate respiratory-illness from energy-poverty channels for cold shocks
- **County unemployment:** replace the ACS proxy with BLS LAUS series
- **Policy counterfactuals:** ACA Medicaid expansion $\times$ climate shocks

Writing and Defence

- **Chapter draft:** unify state, county, and swing narratives into a single empirical chapter
- **Swing robustness:** population-weighted LP profiles for delta results; cumulative IRF plots
- **Thesis write-up:** convert synthesis tables into publication-ready regression tables